PAVAN DHARMA ADAPA

Clinical Data Coordinator · Clinical Data Analyst

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PROFESSIONAL SUMMARY

Clinical data analyst with an M.S. in Computer Science and a year as a graduate research assistant on a public-health study. Manages, validates and documents research data so results hold up to review — and reports what did not work as readily as what did. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Clinical & Health Data: Quality measures (HEDIS, CMS eCQM), value sets, SNOMED CT, LOINC, ICD-10, claims and encounter data, synthetic non-PHI EHR data, HIPAA

Data Management: Data collection and entry workflows, cleaning and standardization, quality control and validation, query resolution, protocol documentation, audit-ready recordkeeping

Analysis & Reporting: SQL, Python (pandas, NumPy, SciPy), R, Advanced Excel, report preparation, Git and version control

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University*, Starkville, MS — Jul 2026 – Present

- Rebuilt the lab's DNA sequencing pipeline into a documented, tested tool that installs in one command — **cutting setup from several days to under an hour** — across 32 merged pull requests covering continuous integration, packaging, and cross-platform support.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus — the verification that made the lab's results defensible to reviewers.
- Wrote the failure diagnostics and the user guide that took the pipeline from **runnable only by its author to runnable by any lab member unaided**, replacing misleading errors with the actual cause.

Graduate Teaching Assistant | *Mississippi State University*, Starkville, MS — Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Graduate Research Assistant | *Mississippi State University*, Starkville, MS — Aug 2024 – Aug 2025

- Managed and analyzed a **900,000+ record** public-health surveillance dataset, and prepared the written and oral reports presented to faculty across two departments.
- Co-author (third of five) on a review manuscript under revision after peer review; contributed the data preparation, statistical modeling and results reporting, and drafted technical sections.
- Built the automated collection and cleaning workflow that standardized multi-source research data, **cutting per-dataset processing from hours to under 10 minutes**.

HEALTH DATA PROJECTS

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Decomposed a diabetes control measure to show **203 of 204 failures were patients never tested**, reframing it as a testing-outreach problem rather than a glycemic-control one.

30-Day Readmission Risk — *UCI Diabetes 130-US Hospitals, de-identified*

- Modeled **99,343 real inpatient encounters**: working the top 20% of the risk list captures **37.7% of all readmissions** at 1.89× enrichment, number-needed-to-screen 4.8.
- Reported AUC 0.664 and argued the ranking — not the AUC — is what a capacity-constrained care-management team consumes.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS — May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India — May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.